

Retail store preliminary BESS assessment

25 September 2026 | Preliminary feasibility assessment | Internal circulation

Recommendation. Carry forward **100-120 kWh usable battery capacity with a 15-20 kW power conversion system (PCS)** for detailed study. This is a screening starting point, not a final design or procurement specification. Continuous refrigeration makes backup resilience and solar integration priorities.

Operating profile and manufacturer reference loads

The store opens **07:00-00:00 (17 hours/day)**. Refrigeration operates **24 hours/day**, with cycling compressors. Lighting operates 17 hours/day.

Installed models remain unidentified. Published input powers below are verified for the named reference models only; they replace the earlier generic ranges as comparison data, not confirmed store loads.

Store equipment	Qty	Reference model	Input per unit
Coca-Cola single-door refrigerator	4	TEFCOLD FSC1450 [1]	265 W
Coca-Cola double-door refrigerator	2	TEFCOLD FS890H BLACK [2]	410 W
Ice-cream freezer	3	TEFCOLD IC402SCE-SO [3]	320 W
Large glass cheese display refrigerator	1	TEFCOLD SOCB25082A [4]	860 W
LED light	40	Site assumption	15 W

TEFCOLD examples are not confirmed Coca-Cola equipment. Frigoglass manufactures Coca-Cola coolers in Egypt with Fresh [5]. Its ICOOL2 ECO800 B page lists **1.09 kWh/day**, but no input wattage [6]. Daily energy is not compressor running power.

Preliminary energy estimate

Reported prepaid electricity-card spending is **EGP 9,000-12,000/month**. The supplied April 2026 commercial tariff assumption is used only as a preliminary estimation basis: EGP 2.79/kWh plus a 3 piaster/kWh consumption stamp (EGP 0.03/kWh), with an EGP 40/month service fee. Tariff applicability and meter deductions require confirmation.

Estimated monthly kWh = (monthly spending - EGP 40) / EGP 2.82/kWh. Daily estimates assume a 30-day month. Prepaid-card spending is not exact monthly consumption: credit balances can carry over between months.

Measure	Low case	Midpoint	High case
Monthly spend (EGP)	9,000	10,500	12,000
Monthly energy (MWh)	~3.2	~3.7	~4.25
Daily energy (kWh/day)	~106	~124	~141
24-hour average load (kW)	~4.4	~5.2	~5.9

Reference-model powers are not a site load schedule and do not validate spending-based consumption. Confirm every installed model, missing loads and actual meter kWh before sizing.

Load composition and engineering implications

Assuming **15 W per LED**, lighting is $40 \times 15 \text{ W} = \mathbf{0.6 \text{ kW}}$ connected load, or **10.2 kWh/day** and **306 kWh/month**. At the midpoint, approximately **3.4 MWh/month (92%)** remains for refrigeration and other loads. Refrigeration therefore appears to dominate the load profile; metering must confirm the split and overnight base load.

BESS options for detailed evaluation

Usable battery energy	Preliminary application
50-70 kWh	Short backup or peak support
80-120 kWh	Strong overnight or partial-day backup
130-180 kWh	Long-duration or near full-day backup

Use **15-20 kW PCS** as the preliminary power range across the options, subject to actual simultaneous load, compressor startup currents, surge capability and phase arrangement. Battery energy sets duration; PCS power sets the load that can be served at a given instant.

At 5.2 kW average, 100-120 kWh gives approximately **19-23 hours before losses and reserve**; this is not guaranteed autonomy. Confirm critical loads, duty cycles, starting charge, temperature and degradation. Define usable capacity at the AC or DC boundary and allow for operating limits when selecting nominal capacity.

Management and economic considerations

A battery alone does not necessarily reduce a flat-tariff bill. Grid charging incurs losses. Evaluate **solar + BESS**, backup resilience and avoided stock loss, or applicable time-of-use and demand/peak-shaving charges. Peak reduction requires a relevant tariff mechanism to save money; no payback is established.

Required next steps

- 1. Measure demand.** Collect 7-14 days of interval metering; establish actual peak kW, overnight base load, compressor cycling and startup currents using suitable high-speed capture for starts.
- 2. Validate consumption.** Read actual monthly kWh from the prepaid meter, reconcile recharge history and balances, confirm tariff/deductions, verify LED wattage and record every machine nameplate.
- 3. Define the duty.** Separate critical and non-critical loads; agree outage duration, autonomy target, acceptable load shedding and required recharge time.
- 4. Assess solar and finalize sizing.** Check roof area, shading and solar potential, then model solar generation, battery dispatch, losses, PCS surge performance and lifecycle economics before final design.

References

Manufacturer sources checked 25 September 2026. References [1]-[4] support comparison-model input powers, not identification of installed equipment. The tariff is a supplied preliminary assumption, not verified by these sources.

[1] TEFCOLD FSC1450 - 265 W

[2] TEFCOLD FS890H BLACK - 410 W

[3] TEFCOLD IC402SCE-SO - 320 W

[4] TEFCOLD SOCB25082A - 860 W

[5] Frigoglass - First commercial cooler produced in Egypt, 12 June 2025

[6] Frigoglass ICOOL2 ECO800 B - 1.09 kWh/day